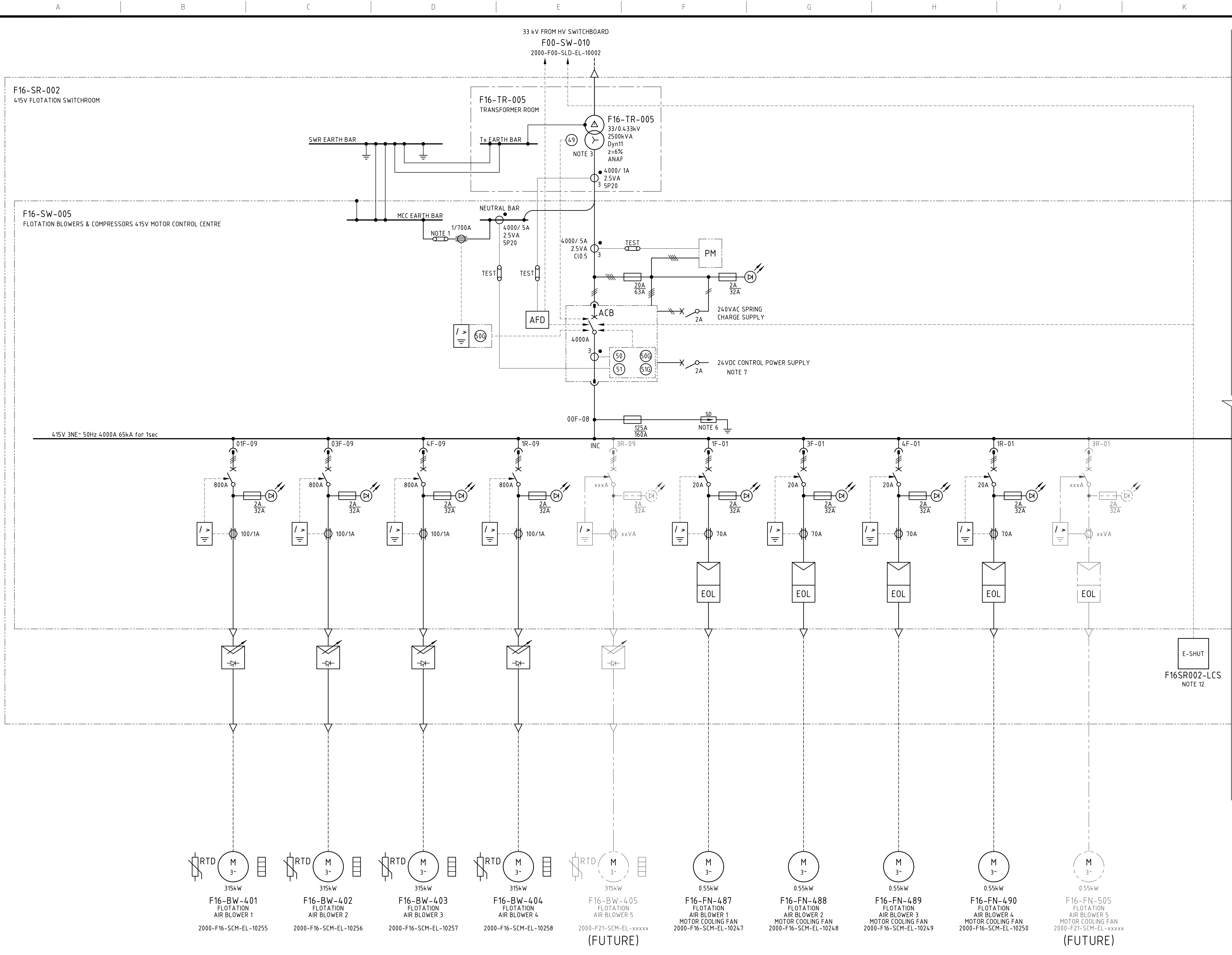


D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10211.DWG 03SEP26 2:29PM ROBIN LAW



CONTINUED ON SHT 2
2000-F16-SLD-EL-10212

NOTES

- EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-005 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
- SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
- UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
- SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
- ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
- SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
- MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
- THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
- TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
- ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
- CIRCUIT BREAKER TO BE SUBSTITUTED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
- EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER
GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER
REVERSING - NOTE 4, 5
- EOL
ELECTRONIC OVERLOAD UNIT
NOTE 4, 5
- THERMAL OVERLOAD UNIT
NOTE 8
- VARIABLE SPEED DRIVE
NOTE 4, 5
- EARTH LEAKAGE RELAY
NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR
(ONLY REQ. FOR MOTORS >37kW - 250kW)
- MOTOR RTD
(ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS
(ONLY REQ. FOR MOTORS >90kW)
- TRANSFORMER PROTECTIVE DEVICES
(REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- ST TRIP SET POINT
CB SIZE
- AFD
ARC FLASH DETECTION
- ECR
EARTH CONTINUITY RELAY
- E-SHUT
SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES	REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2000-F16-SCM-EL-10254	F16-SW-005 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM								
2000-F16-SCM-EL-10253	F16-SW-005 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM								
2000-F16-SCM-EL-10251	F16-SW-005 415V MCC INCOMER SCHEMATIC DIAGRAM								
2000-F16-DRG-EL-10101	F16-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT								
2000-F16-BLK-IC-10001	F16-SR-002 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM								
2000-F16-SLD-EL-10212	F16-SW-005 415V MCC SINGLE LINE DIAGRAM SHEET 2	2	07SEP26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5	1	26AUG25	FAT REDLINES INCORPORATED					
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	11JUN24	APPROVED FOR CONSTRUCTION					



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PRIMERO PROJECT NUMBER: 23517

PROJECT:	NORTHERN STAR RESOURCES LTD KCGM GROWTH PROJECT
TITLE:	ROUGHER AND SCAVENGER FLOTATION F16-SW-005 415V MCC SINGLE LINE DIAGRAM SHEET 1
SHEET	A1
SCALE	NONE
DRG No.	2000-F16-SLD-EL-10211
REVISION	2